

and talented team that has a passion for execution, innovation, collaboration, integrity, and fun. Another key goal is to become an industry-leading technology centre of excellence for Wi-Fi, including Wi-Fi 6.

What do you mean by 'not so simple' wireless, and how does it affect design engineers and the end-product?

When people look at an IoT product, they think it is simply switching a light bulb on or off. So, when I say 'not so simple' wireless, it is important to understand that it is not that simple.

It is tiny but if you break it down, it performs a variety of complex functions. It has a communication function, it performs sense and control, and so on. When you think about the elements of these simple products, they have a lot of the things that a smartphone has today. These products are also becoming intelligent. For instance, when you think of an environment where you can talk to a bulb through Alexa, it needs to detect voice and the keywords, and then start inferencing. When all of these several functions are packed into a tiny ASIC, which has to last on a coin-cell battery, it is not so simple.

You can think of Silicon Labs in the IoT space in the same sense of what smartphones did in innovations to disrupt what happened in the PCs. You will see a lot of these nodes getting more powerful and intelligent over time. As they get more powerful, more and more functions would occur at your node. It is also important to note that when you have this device, which is packed with mixed signal, analogue, RF, along with compute, AI etc, in one ASIC, it is not easy to design.

If you look at a smartphone, the RF chip is separate from the digital chip, and there are various reasons for it. One is obviously the cost benefit, but also because it is difficult to build since analogue tends to be very sensitive to noise, and so on.

That is what I mean when I say 'not-so-simple,' although people only think of the use-case. Turning a light bulb on is not simple when you think about what goes into that function. It is the same case with a lock that has to work for a year on battery life and has to work reliably at the same time. You don't want a lock which is easily hackable, and anybody can open it with a few clicks. So, the security requirements are another big aspect. In fact, in IoT, the security requirements are even more stringent because the more touch

points you have as you connect more devices, the more are the opportunities for attack.

How are you uniquely positioned to disrupt the industry, given your goal to become the Intel of end nodes?

I believe that it is the mindset of working on end nodes where the use cases are very different. Any disruption in the industry does not happen on the flip of a switch, it happens over time. Being in the industry where we are now a pure play IoT company, we have this really unique opportunity to disrupt over time.

Our budget is one tenth of what you may have for a smartphone. As these end nodes slowly become powerful, there will be a time when our ASICs become powerful enough to go into a smartphone but with a much lower power budget, because that's how disruptions occur. You need to start at the bottom. You may think that something is too simple for something to come out of it, but then realise that it has much greater potential that can be tapped into.

So, while bigger chip companies are coming down from higher notes and settling to lower ones, we are tapping the right nodes and trying to broaden the horizon by focusing on only this market, while they instead have to maintain their existing markets.

To a major company, this is a very low performance job that does not align with their business and the technical capabilities that they have. There's no reason why they cannot do what other companies can, but it is incredibly hard for them since they were focused on one thing previously and built that competence over time.

It's not only your revival capabilities but also awareness of the market. By working on these lower end nodes and working with coin-cell battery requirements, you build capabilities over time while learning from your customers. I see it as exciting development in the IoT space, and I believe that you will see a large amount of innovation coming from that space in the next decade.

The use cases will grow, we still have a lot more to catch up because we are focusing on older process nodes. When that happens, we will get enough capability to do even other different things, and that will create new markets or even disrupt existing markets, which is what usually happens. I think this is the law of nature and maybe someday, someone might come and disrupt

Silicon Labs as well. I am excited about what the future holds for us.

Are you trying to replace an entire IoT device with an ASIC?

Yes, right now we are at the end node ASIC stage. Our goal is: What Intel is to PCs, Silicon Labs is to end nodes. We do all the functions to the end nodes, we will do sense control, inferencing. While Amazon Echoes are thin Edge and local data centres are thick Edge, you can think of us as a micro-Edge today in terms of the technical capability of inferencing that we can do today. The idea is to make end nodes more powerful over time. It will not always be centralised where everything goes to a cloud. There are always going to be use-cases, even for latency, cost, privacy, security, that require decision making, and we wish to be that—the Intel of end nodes.

How do you see the open source phenomena taking place in the chip design and component design space? Do you see that as an enabler or a threat where it may just disrupt the game totally?

I personally view it as an enabler. It's always good to have a competition, and I think it makes you better. I would say that open source has unique advantages and disadvantages.

If you're a business, you have to figure out how to make money because that's the main priority. But there are people who are recognising that there are techniques and mechanisms even with open source, whether in hardware or software, to drive your business and drive differentiation. If you can drive differentiation and value to the customer, people will adapt even if you are using an open source architecture. All it means is that there are different licensing terms when compared to others.

You still have to build products, and that is just a component that has to sit inside your product. A processor by itself cannot do much if your memory architecture is not good enough. I have seen that whether you licence IP or not, there are still opportunities for you to innovate because the system is so complex. They all have to work together, and you have to make it better over time. There are always improvements that are going on. I pursue it more as an opportunity, even though it is always a threat. My mindset is to find out how to convert it into an opportunity because it's a revolution that you cannot stop. I think that all of us have to embrace it and convert that into an opportunity. **EFY**